Stochastic Pit Stop Optimisation

via Markov Decision Processes, Gaussian Process Tyre Degradation,

and Bayesian Safety Car Modelling

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We formulate the Formula 1 pit stop timing decision as a finite-horizon **Markov Decision Process** (MDP) and solve it exactly via backward induction [1]. The transition kernel integrates two stochastic components: a Gaussian Process regression model with Matérn 5/2 kernel for tyre degradation — returning a full posterior distribution over lap time delta — and a Non-Homogeneous Poisson Process with Bayesian Beta-Binomial conjugate updating for safety car deployment probability. Monte Carlo simulation with antithetic variates evaluates the policy over stochastic race trajectories. Compared with 57 real driver strategies at three 2023 circuits using modelled time deltas, the policy is faster in **53%** of comparisons, by a mean of **0.2 seconds**, with results differing sharply by circuit. In the saved Monte Carlo run at Silverstone it is 48 seconds slower on average than a baseline heuristic, so these results do not yet demonstrate a strategic advantage. We further establish an exact isomorphism between this framework and the American option optimal exercise problem (Snell envelope), demonstrating that pit stop timing and early option exercise are the same mathematical problem in different physical domains.

Keywords: Markov Decision Process · Dynamic Programming · Gaussian Process Regression · Bayesian Inference · Monte Carlo Simulation · Optimal Stopping · Formula 1 Strategy

1. Introduction

Every lap of every Formula 1 race, a pit wall strategist faces a sequential decision under uncertainty: pit now, or stay out? The naive formulation—"pit when the tyres are worn"—fails to account for the stochastic structure of the problem. Tyre degradation is non-linear, compound-specific, and accelerates sharply past a critical age (the *cliff*). Safety car deployment is random and lap-varying. The cumulative effect of all future decisions must be considered simultaneously.

This paper treats the pit stop decision as a *finite-horizon MDP* — the canonical mathematical framework for sequential decision problems under uncertainty. The exact backward-induction solver guarantees globally optimal policy recovery. We contribute:

1. A Weibull survival + Gaussian Process hybrid tyre model;
2. A Bayesian NHPP safety car model with closed-form conjugate updates;
3. An exact isomorphism between pit stop timing and American option exercise;
4. An open-source implementation with a partial comparison against 2023 strategies at three circuits.

2. MDP Formulation

2.1 Tuple Definition

Definition 2.1 (Pit Stop MDP)**.** The race is a finite-horizon MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, T)$ where T is the total number of laps and a decision is made at each lap $t \in \{1, \ldots, T\}$.

2.2 State Space

The state vector s must be Markovian: the future is independent of the past given the present. We define:

$$s = (t,\ c,\ a,\ p,\ g_{\text{ahead}},\ g_{\text{behind}},\ n_{\text{pit}},\ \not\Vdash_{\text{SC}}) \tag{2.1}$$

where $c \in \{\text{S}, \text{M}, \text{H}, \text{I}, \text{W}\}$ is the tyre compound, $a \in \{0, \ldots, 50\}$ is tyre age in laps, $p \in \{1, \ldots, 20\}$ is race position, $g_{\text{ahead}}, g_{\text{behind}} \in \{0, 1, 2, 3\}$ are discretised gap bins, $n_{\text{pit}} \in \{0, 1, 2\}$ is pit stops taken, and $\not\Vdash_{\text{SC}}$ indicates safety car status. After pruning unreachable states, approximately 13 million states remain.

2.3 Action Space

$\mathcal{A}(s) \subseteq \{\text{STAY}, \text{PIT-S}, \text{PIT-M}, \text{PIT-H}\}$. Pit actions are unavailable when $n_{\text{pit}} = 2$. STAY is unavailable at lap T if $n_{\text{pit}} = 0$ (FIA mandatory stop regulation, Article 30.5).

2.4 Bellman Optimality Equation

Proposition 2.1 (Optimal Value Function)**.** The optimal value function $V^*(s, t)$ satisfies:

$$V^*(s,t) = \max_{a \in \mathcal{A}(s)} \Big[R(s,a) + \sum_{s'} P(s' \mid s, a)\, V^*(s', t{+}1) \Big] \tag{2.2}$$

with boundary condition $V^*(s, T+1) = 0$ for all s.

The optimal policy is recovered as:

$$\pi^*(s,t) = \arg\max_{a \in \mathcal{A}(s)} Q^*(s,a,t) \tag{2.3}$$

The backward-induction solver (Algorithm 1) computes $V^*(s, t)$ exactly for all states in $O(T \cdot |\mathcal{S}| \cdot |\mathcal{A}|)$ time. Bellman's principle of optimality guarantees global optimality—not a local or approximate solution.

2.5 Reward Function

The reward represents negative time cost relative to the theoretical minimum:

$$R(s, \text{STAY}) = -\text{deg}(a, c, \not\Vdash_{\text{SC}}) \tag{2.4}$$

$$R(s, \text{PIT-}c) = -\Delta_{\text{pit}}(\text{circuit}) \tag{2.5}$$

where $\Delta_{\text{pit}} \in [18.2\text{s}, 22.0\text{s}]$ is the circuit-specific pit lane time penalty.

```python
# Boundary condition
V[T+1] = defaultdict(float)  # V*(s,T+1)=0

# Backward sweep
for t in range(T, 0, -1):
  for s in reachable_states(t):
    V[t][s] = max(
      R(s,a) + sum(
        P(s_,s,a) * V[t+1][s_]
        for s_, p in transition(s,a)
      )
      for a in A(s)
    )
```

Listing 1: Backward induction core loop (solver.py)

3. Tyre Degradation Model

3.1 Weibull Survival Model

The Weibull distribution is the workhorse of reliability engineering, modelling time-to-failure of physical systems. We model cumulative lap time delta $D(a)$ at tyre age a as:

$$D(a) = D_0 \left(1 - e^{-(a/\lambda)^k}\right) \tag{3.1}$$

where $k > 1$ is the shape parameter (accelerating degradation implies $k > 1$), λ is the scale parameter (characteristic life in laps), and D_0 is the maximum pace loss. The instantaneous degradation rate is:

$$\frac{dD}{da} = D_0 \cdot \frac{k}{\lambda} \left(\frac{a}{\lambda}\right)^{k-1} e^{-(a/\lambda)^k} \tag{3.2}$$

The *cliff lap*—the inflection point where degradation rate is maximised—is:

$$a_{\text{cliff}} = \lambda \left(\frac{k-1}{k}\right)^{1/k} \tag{3.3}$$

This corresponds to $d^2D/da^2 = 0$, i.e. the lap at which degradation begins to accelerate most rapidly.

3.2 Fitted Parameters (2023 Season)

Parameters are fitted via non-linear least squares on clean-air laps (gap_ahead > 2s, green flag) from the 2021–2023 FastF1 dataset. Table 1 reports results.

The negative R^2 values mean that the fitted curves explain less variance than a constant, and several parameters sit at their bounds. The parametric fit alone is therefore a weak description of the data, and no clear cliff is identified.

Table 1: Weibull parameters fitted to 2021–2023 FastF1 telemetry.

Compound	k	λ	D_0 (s)	a_{cliff}	R^2
SOFT	1.00	5.0	0.77	0	−0.162
MEDIUM	1.00	60.0	1.61	0	−0.067
HARD	3.55	60.0	1.79	≈ 55	−0.057

3.3 Gaussian Process Regression

The Weibull model is parametric but cannot capture circuit-specific deviations. We augment it with a Gaussian Process fitted on the residuals:

$$f(a) = \mu_{\text{Weibull}}(a) + \epsilon(a), \quad \epsilon \sim \mathcal{GP}(0,\, k_{\text{Matérn}}(a, a')) \tag{3.4}$$

We use the Matérn $\frac{5}{2}$ kernel, which is $\mathcal{C}^2$ differentiable—smooth enough for physical wear, flexible enough for the cliff:

$$k(a, a') = \sigma^2 \left(1 + \frac{\sqrt{5}\, r}{\ell} + \frac{5r^2}{3\ell^2}\right) e^{-\frac{\sqrt{5}\, r}{\ell}}, \quad r = |a - a'| \tag{3.5}$$

The GPR posterior yields $(\mu^*(a), \sigma^*(a))$. For risk-averse strategy, we use the upper confidence bound:

$$\text{UCB}(a) = \mu^*(a) + \beta\, \sigma^*(a), \quad \beta \in [1, 2] \tag{3.6}$$

This is formally equivalent to UCB exploration in multi-armed bandits [8]. β controls the pit-early vs. stay-out-longer tradeoff, just as it controls exploration vs. exploitation in bandits.

4. Safety Car Model

4.1 Non-Homogeneous Poisson Process

Safety car (SC) deployment is modelled as an NHPP with lap-varying intensity $\lambda(t)$. Lap 1 has approximately 4× the baseline rate; the rate decays to baseline by lap 5 and remains roughly constant through the mid-race, with a slight decrease in the final third as the field spreads.

4.2 Bayesian Beta-Binomial Updating

For each lap t, we maintain a Beta posterior over the deployment probability p_t:

$$p_t \sim \text{Beta}(\alpha, \beta) \quad \text{(prior)} \tag{4.1}$$

$$k_t \mid p_t \sim \text{Binomial}(n_{\text{races}}, p_t) \quad \text{(likelihood)} \tag{4.2}$$

$$p_t \mid k_t \sim \text{Beta}(\alpha + k_t,\ \beta + n - k_t) \tag{4.3}$$

The Beta-Binomial conjugacy yields a closed-form posterior—no MCMC required. The posterior mean $\mathbb{E}[p_t \mid k_t] = (\alpha + k_t)/(\alpha + \beta + n)$ is our point estimate; the variance provides uncertainty bounds.

On held-out test data, the saved run gives an MAE of 4.05 percentage points, and the 90% credible intervals contain the observed frequency in only 8% of laps. The intervals are therefore far too narrow and the model is not yet well calibrated.

4.3 Duration: Geometric Distribution

Once deployed, SC duration is modelled as Geometric(p_{clear}):

$$\mathbb{P}(\text{duration} = d) = p_{\text{clear}}(1 - p_{\text{clear}})^{d-1} \tag{4.4}$$

MLE from the saved data: $\hat{p}_{\text{clear}} \approx 0.42$ (mean duration ≈ 2.4 laps). A Kolmogorov-Smirnov test fails to reject the Geometric fit ($p = 0.31$).

4.4 The Safety Car as an American Option

Under SC, the pit lane delta collapses from ≈21s to ≈6s. The expected value of this window at lap t with k laps remaining:

$$V_{\text{SC}}(t) = \mathbb{P}(\text{SC} \in [t, t{+}k]) \cdot (\Delta_{\text{green}} - \Delta_{\text{SC}}) \tag{4.5}$$

This decays as $k \to 0$, exactly like American option time value. The Bellman equation accounts for this implicitly: the solver correctly prefers to hold the option when SC probability is high.

5. Isomorphism with Quantitative Finance

Proposition 5.1 (MDP–Snell Envelope Isomorphism)**.** The Bellman equation (2.2) is structurally identical to the Snell envelope [5] for American option optimal exercise:

$$U_t = \max\{Z_t,\ \mathbb{E}[U_{t+1} \mid \mathcal{F}_t]\} \tag{5.1}$$

where Z_t is the exercise value at time t and U_t is the Snell envelope. Identifying $Z_t \leftrightarrow R(s, a_{\text{pit}})$ and $\mathbb{E}[U_{t+1}] \leftrightarrow \sum_{s'} P(s'|s, a)\, V^*(s', t{+}1)$ gives the exact correspondence.

Remark 5.1. This is not a metaphor. Table 2 shows the exact structural mapping. The solver in `solver.py` solves both problems; only $R(s, a)$ and $P(s'|s, a)$ differ.

Table 2: Exact isomorphism: F1 strategy ↔ quant finance.

F1 Strategy	Quantitative Finance
Pit stop timing	American option exercise
Tyre degradation	Theta (time value decay)
Safety car window	Volatility event
Undercut threat	Early exercise premium
Bellman equation	Snell envelope
UCB risk-averse pit	VaR / CVaR constraint
Monte Carlo races	Path-dependent pricing
Beta-Binomial SC model	Bayesian jump process

6. Monte Carlo Validation

6.1 Simulation Design

We evaluate the optimal policy via Monte Carlo race simulation. The simulator defaults to $N = 10{,}000$ runs; the saved results below use $N = 2{,}000$ at Silverstone. Each simulation independently samples SC events from the calibrated NHPP, tyre degradation noise from the GPR posterior, and gap dynamics from an empirical transition model.

6.2 Antithetic Variates

For each random seed u, we evaluate both the primary trajectory and its antithetic $(1-u)$. The antithetic pair has negative correlation, reducing the estimator variance:

$$\mathrm{Var}\left[\frac{f(u)+f(1-u)}{2}\right] = \frac{\mathrm{Var}(f)+\mathrm{Cov}(f(u),f(1-u))}{2} < \frac{\mathrm{Var}(f)}{2} \tag{6.1}$$

when f is monotone in u.

6.3 Results

Table 3: Monte Carlo results at Silverstone ($N = 2{,}000$): MDP policy vs baseline heuristic.

Policy	Mean race time (min)
MDP policy	81.73
Baseline heuristic	80.93

The MDP policy is 48.1 seconds *slower* on average than the baseline, with a 5% lower standard deviation of race time. This run does not show that the policy improves on the heuristic, and it covers one circuit only.

7. 2023 Season Backtest

We apply the optimal policy to actual 2023 race data with real SC timing as ground truth. For each race, we compare our recommended pit lap to the actual pit lap executed by each team, estimating the time delta:

$$\Delta t = (a_{\text{actual}} - a_{\text{MDP}}) \cdot \left.\frac{dD}{da}\right|_{a=\bar{a}} \tag{7.1}$$

Across 3 circuits ($N = 57$ driver comparisons, `results/backtest_summary.csv`), the MDP policy is faster in **53%** of comparisons. The mean delta is **+0.20 seconds** overall, +0.87 seconds when counted as faster and −0.54 seconds otherwise. By circuit, it is faster in 85% of comparisons at Monaco (mean +1.01 s), 72% at Silverstone (+0.08 s) and 0% at Monza (−0.54 s).

These are modelled deltas, not observed race times. Each circuit uses a single recommended lap, computed for a medium-tyre car in fifth place, and the comparison uses each driver's first stop only. The results are indicative at most.

Limitations. Undercut dynamics and rival reactions are not modelled, nor are tyre allocation constraints or weather transitions. Driver, car and compound differences are not reflected in the single per-circuit recommendation.

8. Conclusion

We have presented PITWALL: a complete MDP-based framework for F1 pit stop optimisation with exact backward-induction solver, hybrid Weibull + GPR tyre model, Bayesian NHPP safety car model, and formal isomorphism with American option exercise. The evaluation is mixed: a small modelled gain against real strategies that varies by circuit, and a 48-second deficit against a baseline heuristic in simulation. Demonstrating strategic value would require a better tyre fit and a full-season backtest on observed race outcomes.

Future work includes multi-agent formulations (rival team reactions as Nash equilibria), weather uncertainty integration, and extension to tyre allocation optimisation across a full race weekend.

Full implementation, test suite, and interactive simulator: `claudialbombin.github.io/pitwall`.

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